



### REVA Academy for Corporate Excellence

|                                               |                                       |
|-----------------------------------------------|---------------------------------------|
| Program: MTech/MSc in Artificial Intelligence | Course: Programming Essentials for AI |
| Batch: AI09                                   | Exam: Final                           |
| Date: 14 <sup>th</sup> Dec 2024               | Time: 02.00 pm to 5.00 pm             |
| Duration: 3 Hrs                               | Marks: 100                            |

#### Instructions

1. **Code Comments:** Add necessary comments throughout your code to explain each step clearly. This will help in understanding your thought process and make your code more readable.
2. **Integrity:** Do not cheat. Complete the exam using your own understanding and skills.
3. **Submission:** Submit the Jupyter Notebook (.ipynb file) with the questions numbered in order.
4. **DON'T CHANGE THE ORDER OF QUESTIONS.** Make sure the code, comments, and explanations are clear and properly formatted.
5. Personal laptops are not allowed.

#### Section A

|     |                                                                                                                                                                                                                                                                                 |   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q1. | <p>What will be the missing code ____ to generate the following output?</p> <p>Output-</p> <pre>[20, 30, 40, 50]</pre> <p>Code-</p> <pre>my_list = [10, 20, 30, 40, 50] result = my_list[1:4] + [____] print(result)</pre>                                                      | 5 |
| Q2. | <p>What will be the missing code ____ to generate the following output?</p> <p>Output-</p> <pre>Hello, W_o_r_l_d!</pre> <p>Code-</p> <pre>greeting = "Hello" target = "World" formatted_target = " ".join(____) result = f"{greeting}, {formatted_target}!" print(result)</pre> | 5 |
| Q3. | <p>What will be the missing code ____ to generate the following output?</p> <p>Output-</p>                                                                                                                                                                                      | 5 |

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|     |                                                                                                                                                                                                                                                             |   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|     | <p>Caught an exception: division by zero<br/>Execution complete</p> <p>Code-</p> <pre>try:     x = 10 / 0 except ____ as e:     print(f"Caught an exception: {e}") finally:     print("Execution complete")</pre>                                           |   |
| Q4. | <p>What will be the missing code ____ to generate the following output?</p> <p>Output-</p> <pre>[2, 4]</pre> <p>Code –</p> <pre>numbers = [1, 2, 3, 4, 5] result = [num for num in numbers if ____] print(result)</pre>                                     | 5 |
| Q5. | <p>What will be the missing code ____ to generate the following output?</p> <p>Output-</p> <pre>[4 5 6]</pre> <p>Code-</p> <pre>import numpy as np  array = np.array([[1, 2, 3], [4, 5, 6]]) result = array[1, ____] print(result)</pre>                    | 5 |
| Q6. | Write a function named find_largest(numbers) that takes a list of integers and returns the largest number in the list.                                                                                                                                      | 5 |
| Q7. | <p>Create a class named Rectangle with the following:</p> <ul style="list-style-type: none"> <li>An __init__ method to initialize the length and width attributes.</li> <li>A method named calculate_area that returns the area of the rectangle</li> </ul> | 5 |



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|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q8.  | Write a program that uses a while loop to calculate the sum of all numbers from 1 to 10. Use the result to generate the exact output shown above.                                                                                                                                                                                                                                                                                                  | 5 |
| Q9.  | <p>Create a Python program that accepts a series of numbers from the user and calculates their product. The program should:</p> <ul style="list-style-type: none"> <li>• Prompt the user to enter numbers one by one.</li> <li>• Stop taking inputs when the user types "done".</li> <li>• Calculate and display the product of all entered numbers.</li> </ul>                                                                                    | 5 |
| Q10. | <p>Create a Python program to parse the JSON data and extract specific information. The program should parse the following JSON string, which contains information about students, and display the name of the first student.</p> <pre>{   "students": [     {"name": "Alice", "age": 22, "major": "Computer Science"},     {"name": "Bob", "age": 24, "major": "Physics"},     {"name": "Charlie", "age": 21, "major": "Mathematics"}   ] }</pre> | 5 |

#### Section B: All questions are mandatory.

|              |                                                                                                                                                                                                                                                                                      |       |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Instructions | <p>Case Study: Titanic Dataset Analysis</p> <ul style="list-style-type: none"> <li>• Load the Titanic dataset from the Seaborn library.</li> <li>• Use Pandas to answer the following questions and present the results using appropriate charts in Matplotlib or Seaborn</li> </ul> | Marks |
| Q11.         | What was the total number of survivors in the third class (Pclass = 3)?                                                                                                                                                                                                              | 6     |
| Q12.         | What was the average age of female passengers who survived?                                                                                                                                                                                                                          | 6     |
| Q13.         | What was the survival rate of passengers aged 0-18 years?                                                                                                                                                                                                                            | 6     |
| Q14.         | What was the survival rate of male passengers traveling alone (i.e., sibsp == 0 and parch == 0)?                                                                                                                                                                                     | 6     |



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|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q15. | What was the survival rate of passengers with siblings/spouses aboard (i.e., sibsp > 0)?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6  |
| Q16. | <p>Conduct an exploratory data analysis (EDA) on the Titanic dataset to uncover meaningful insights. Your analysis should include:</p> <ol style="list-style-type: none"> <li>Summary statistics and data distribution for numerical and categorical variables.</li> <li>Analysis of missing values and their potential impact.</li> <li>Visualization of key relationships (e.g., survival rate by class, age, gender, etc.).</li> <li>Identification of potential outliers or anomalies.</li> <li>A summary of key insights derived from your analysis, focusing on factors influencing survival.</li> </ol> | 20 |

\*\*\*\*\*End\*\*\*\*\*